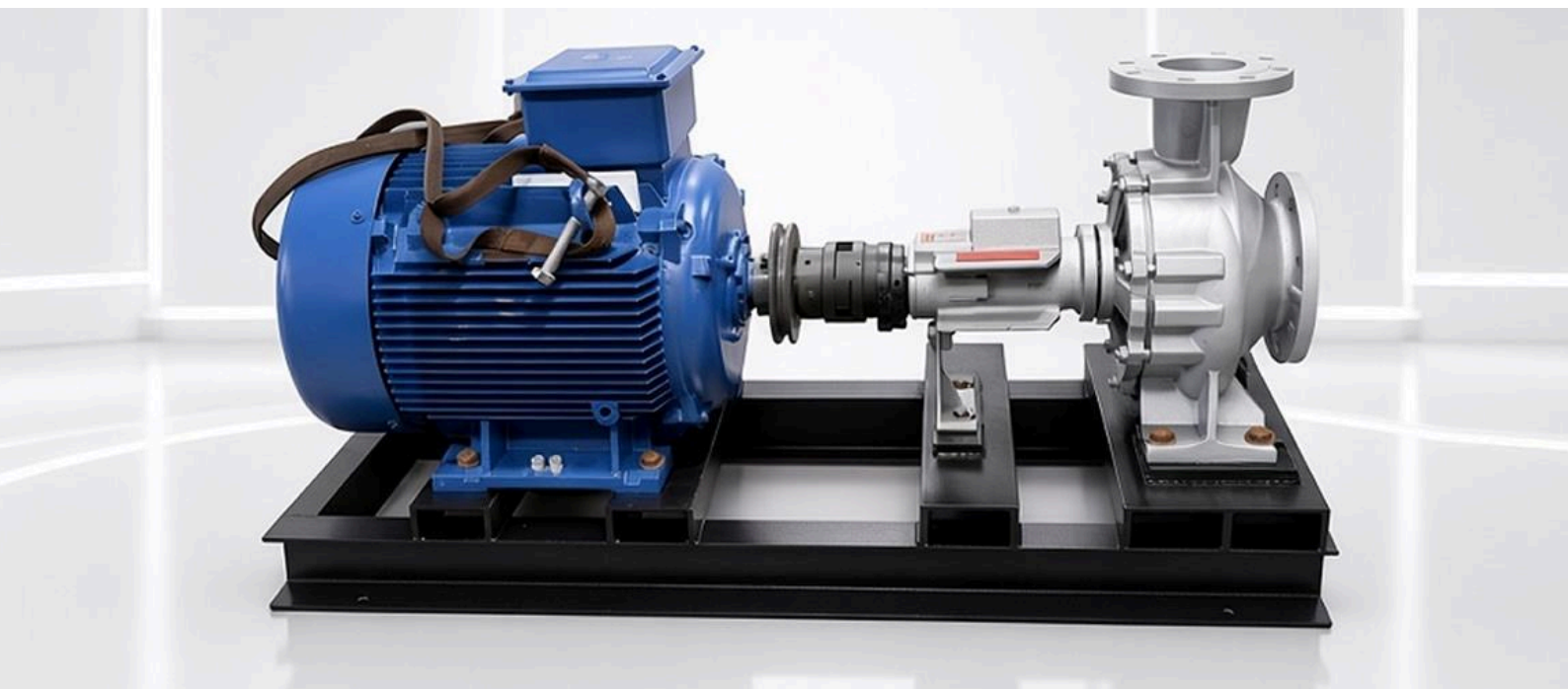




Product Guide: Advantages · Selection Table · Model Datasheets · Competitor Comparison

Redefine Pumping

Founded 1997 · 28+ Years of Excellence · DIN 24256 / ISO 2858 · Made in India

SECTION 1 — iPump TECHNICAL OVERVIEW

What is the iPump?

The K-Fins **iPump** (ATP Series) is a horizontal, single-stage, end-suction, back pull-out centrifugal pump designed specifically for thermic fluid (thermal oil) and hot water circulation. Manufactured to **DIN 24256 / ISO 2858** standard, it requires **no external water cooling or air conditioning** — the finned distance piece (heat barrier) dissipates heat by natural air convection. Ideal for temperatures up to **350°C**.

Parameter	Thermal Fluid	Hot Water
Max. Flow Rate	10 – 750 m ³ /h	10 – 750 m ³ /h
Max. Head	8 – 110 metres	8 – 110 metres
Fluid Temperature	Up to 350°C	80°C – 180°C
Max. Operating Pressure	16 bar	16 bar
Speed	1450 / 2900 RPM	1450 / 2900 RPM
Standard	DIN 24256 / ISO 2858	DIN 24256 / ISO 2858

Materials of Construction

Component	Material	Component	Material
Volute Casing	Nodular Cast Iron GJS-1025	Impeller	Nodular Cast Iron GJS-1025
Discharge Cover	Nodular Cast Iron GJS-400-15	Wear Rings	Grey Cast Iron
Shaft	Chrome Steel 1.4021 + OTHRC55	Bearing Bracket	Nodular Cast Iron GG-26

* Alternative materials available on request. ICP Series available in SS 304, SS 316, SS 410, SS 420, Noble Alloys.

SECTION 2 — iPump KEY ADVANTAGES

No.	Advantage	Technical Benefit	Customer Value
1	No External Cooling Required Air-cooled finned distance piece dissipates heat by natural convection	Eliminates cooling water piping, pumps, heat exchangers, and associated maintenance	Lower installation cost. Zero water consumption. Works in water-scarce locations.
2	DIN 24256 / ISO 2858 Standard International standard back pull-out design	Dimensional interchangeability with any other DIN/ISO standard pump worldwide	Easy replacement of competitor pumps without piping modifications.
3	Back Pull-Out Design Rotating assembly removable without disturbing pipework	Impeller, seal, shaft and bearings accessible for inspection without unbolting any pipe flange	Minimum downtime during maintenance. No pipe alignment required after service.
4	Operates up to 350°C Designed for high-temperature thermic fluid service	Suitable for all thermic fluid heater applications — textile, plywood, chemical, food industries	One pump covers the full temperature range of industrial thermic fluid systems.
5	Optimised Heat Barrier Finned distance piece with 110mm+ barrier length	Keeps bearing housing and seal cavity at safe operating temperature without any external cooling medium	Extended bearing and seal life. Reduced bearing failure risk at high temperatures.
6	Mechanical Seal — Tandem Arrangement Available with EagleBurgmann seals	Double seal arrangement prevents any thermal oil leakage to atmosphere — meets safety and fire prevention requirements	Safe operation with flammable thermal oils. Meets plant safety and insurance requirements.
7	Reinforced Deep-Groove Ball Bearings Special high-temperature grease fill	Designed to handle both radial and axial thrust loads at elevated temperatures for extended service life	Longer bearing life. Fewer bearing replacements. Lower maintenance cost.
8	Optimised Venting Contour Highly effective venting design built into casing	Ensures complete air/vapour removal during startup — prevents dry running and cavitation during initial filling	Reliable startup every time. Prevents impeller damage from vapour-locked operation.
9	Variable Speed Compatible Designed for VFD/variable speed drive operation	Flow can be controlled by speed variation instead of throttling — reduces energy consumption significantly	Energy savings up to 30-50% with VFD. Reduces control valve wear.
10	Made in India — Fast Delivery & Local Support Manufactured at Pandesara GIDC, Surat since 1997	All spare parts locally available in India. No import lead times. Technical support by experienced engineers.	3-5x faster spare parts delivery vs imported brands. Lower total cost of ownership.
11	Impeller Trimmed to Duty Point Custom impeller trim per order	Impeller diameter trimmed to match exact specified flow and head — ensures operation at or near BEP	Maximum efficiency at actual operating conditions. Lower motor load. Energy saving.
12	Confined Gaskets — Zero Blow-Out Risk Gaskets fully confined on all sides	Gasket cannot extrude or blow out under operating pressure — even at elevated temperatures	Safe operation with hot oils. Prevents dangerous leaks at high-temperature flanged joints.

SECTION 3 — iPump SELECTION GUIDE

How to Use This Guide: Identify your required Flow (m³/h) and Head (m). Match to the model range below. For 1450 RPM: preferred for high-temperature, viscous, or seal-sensitive duties. For 2900 RPM: preferred for higher head requirements with lower flow. Contact K-Fins for a duty-specific selection with trimmed impeller.

Step 1 — Quick Application Selection

Application	Fluid	Temp. Range	Recommended Speed	Seal Arrangement
Thermic Fluid Heater Circulation	Thermal Oil (e.g. Therminol, Dowtherm)	Up to 350°C	1450 RPM	Mechanical Seal — Tandem (API Plan 52)
Hot Water Heating System	Hot Water	80°C – 180°C	1450 or 2900 RPM	Single Mechanical Seal
Coal Tar / Pitch Circulation	Coal Tar Oil (cokable fluid)	Up to 280°C	1450 RPM preferred	API Plan 32 + Plan 62 (hard SiC/SiC)
Bitumen / Asphalt Handling	Bitumen, modified bitumen	Up to 200°C	1450 RPM	Jacketed casing + seal flush
Chemical Process (hot liquid)	Process chemicals, solvents	Up to 200°C	1450 or 2900 RPM	Per fluid compatibility

Step 2 — Model Selection Table at 1450 RPM

Models follow DN Suction × Impeller Diameter (mm) naming convention per DIN 24256 / ISO 2858. Flow and head values shown are approximate maximum at full impeller diameter. Actual duty point depends on impeller trim — contact K-Fins for exact selection.

Model	Max Flow (m ³ /h)	Max Head (m)	Typical Motor (kW)	Best For
32-125	5 – 12	8 – 15	0.75 – 1.5	Small thermic fluid lines, lab applications
32-160	5 – 12	15 – 22	1.5 – 2.2	Small heater circuits
32-200	5 – 12	20 – 30	2.2 – 3.7	Small industrial heaters
32-250	5 – 12	28 – 40	3.7 – 5.5	Extended circuit small heaters
40-125	8 – 20	8 – 15	0.75 – 2.2	Textile stenter thermic fluid
40-160	8 – 20	14 – 22	1.5 – 3.7	Textile / plywood heaters
40-200	8 – 20	20 – 32	2.2 – 5.5	Medium textile / chemical plant
40-250	8 – 20	28 – 45	3.7 – 7.5	Higher head medium flow circuits
40-315	8 – 20	38 – 55	5.5 – 11	Extended circuit medium heaters
50-125	15 – 36	8 – 15	1.5 – 3.7	Textile / paper mill thermic fluid
50-160	15 – 36	14 – 22	2.2 – 5.5	Medium thermic fluid heater
50-200	15 – 36	20 – 35	3.7 – 7.5	Standard plywood / textile plant
50-250	15 – 36	28 – 48	5.5 – 11	Higher head medium circuits
50-315	15 – 36	38 – 60	7.5 – 15	Large plywood / rubber plants
65-125	25 – 60	8 – 15	2.2 – 5.5	Large textile stenter lines
65-160	25 – 60	14 – 22	3.7 – 7.5	Standard thermic fluid heater
65-200	25 – 60	20 – 35	5.5 – 11	Sugar / chemical process heaters
65-250	25 – 60	28 – 50	7.5 – 15	Large textile / paper plants
65-315	25 – 60	38 – 65	11 – 22	Extended high-head circuits

Model	Max Flow (m³/h)	Max Head (m)	Typical Motor (kW)	Best For
80-160	40 – 90	14 – 22	5.5 – 11	Chemical / pharma process heaters
80-200	40 – 90	20 – 35	7.5 – 15	Large thermic fluid heater
80-250	40 – 90	28 – 50	11 – 22	Large industrial plants
80-315	40 – 90	38 – 65	15 – 30	High-head large circuits
80-400	40 – 90	55 – 80	22 – 45	Very high head large plants
100-160	60 – 140	14 – 22	7.5 – 18.5	Large chemical / refinery heaters
100-200	60 – 140	20 – 35	11 – 22	Standard large industrial heater
100-250	60 – 140	28 – 50	15 – 30	Large plant thermic fluid system
100-315	60 – 140	38 – 65	22 – 45	High-head large industrial circuits
100-400	60 – 140	55 – 85	30 – 55	Very high head large plants
125-200	90 – 220	20 – 35	15 – 30	Oil & gas / large chemical plant
125-250	90 – 220	28 – 50	22 – 45	Large oil & gas heater circuits
125-315	90 – 220	38 – 65	30 – 55	Large process plant thermic fluid
125-400	90 – 220	55 – 85	45 – 75	Very large industrial plant
150-200	150 – 350	20 – 35	22 – 45	Very large process heater
150-250	150 – 350	28 – 50	30 – 55	Large oil & gas plant
150-315	150 – 350	38 – 65	45 – 90	Major industrial thermic fluid plant
150-400	150 – 350	55 – 90	55 – 110	Largest industrial plants

Note: Flow and head values are approximate indicative ranges at full impeller diameter. Actual performance depends on system curve, impeller trim, fluid viscosity and specific gravity. Always request a formal pump selection from K-Fins engineering team for final specification.

Step 3 — Model Selection Table at 2900 RPM

2900 RPM models are available for 32 to 125 suction bore sizes. Preferred where higher head is required at moderate flow. Not recommended for high-temperature duties above 300°C without bearing assessment.

Model	Max Flow (m³/h)	Max Head (m)	Typical Motor (kW)	Best For
32-125	5 – 12	20 – 35	1.5 – 3.7	Small high-head thermic circuits
32-160	5 – 12	35 – 55	2.2 – 5.5	Compact high-head systems
32-200	5 – 12	50 – 75	3.7 – 7.5	High head small flow circuits
32-250	5 – 12	65 – 95	5.5 – 11	Very high head small flow
40-125	8 – 20	20 – 35	2.2 – 5.5	Textile / compact systems
40-160	8 – 20	35 – 55	3.7 – 7.5	Medium flow high head
40-200	8 – 20	50 – 75	5.5 – 11	Chemical process high head
40-250	8 – 20	65 – 95	7.5 – 15	High head medium flow circuits
50-125	15 – 36	20 – 35	3.7 – 7.5	Textile stenter high head
50-160	15 – 36	35 – 55	5.5 – 11	Medium flow high head thermic
50-200	15 – 36	50 – 75	7.5 – 15	Large stenter / plywood high head
50-250	15 – 36	65 – 95	11 – 22	High head medium circuits
65-125	25 – 60	20 – 35	5.5 – 11	Large flow moderate head
65-160	25 – 60	35 – 55	7.5 – 15	Large thermic fluid high head

Model	Max Flow (m³/h)	Max Head (m)	Typical Motor (kW)	Best For
65-200	25 – 60	50 – 75	11 – 22	Chemical / sugar plant
65-250	25 – 60	65 – 95	15 – 30	High head large flow
80-160	40 – 90	35 – 55	11 – 22	Large plant high head duty
80-200	40 – 90	50 – 75	15 – 30	Large industrial high head
80-250	40 – 90	65 – 95	22 – 45	Oil & gas high head duty
100-160	60 – 140	35 – 55	15 – 30	Large chemical / refinery duty
100-200	60 – 140	50 – 75	22 – 45	Large industrial high head
100-250	60 – 140	65 – 95	30 – 55	Major plant high head circuit
125-200	90 – 220	50 – 75	30 – 55	Very large plant high head

Note: Performance curves for all individual models are available from K-Fins engineering. Values shown are indicative at full impeller diameter. Contact K-Fins for certified test data.

SECTION 4 — MODEL-WISE QUICK DATASHEETS

Note: Data below is based on K-Fins catalogue and indicative performance values. All performance curves should be verified with works test data before contractual commitment. Contact K-Fins for duty-specific selection with trimmed impeller and motor recommendation.

iPump 32-125 @ 1450 RPM			iPump 50-200 @ 1450 RPM		
Flow Range	5 – 12 m³/h	Head Range	Flow Range	15 – 36 m³/h	Head Range
Motor Range	0.75 – 1.5 kW	Impeller Dia.	Motor Range	3.7 – 7.5 kW	Impeller Dia.
Std. Seal	Mechanical seal / Gland packing	Application	Std. Seal	Mechanical seal	Application
iPump 65-200 @ 1450 RPM			iPump 80-250 @ 1450 RPM		
Flow Range	25 – 60 m³/h	Head Range	Flow Range	40 – 90 m³/h	Head Range
Motor Range	5.5 – 11 kW	Impeller Dia.	Motor Range	11 – 22 kW	Impeller Dia.
Std. Seal	Mechanical seal	Application	Std. Seal	Mechanical seal	Application
iPump 100-250 @ 1450 RPM			iPump 125-250 @ 1450 RPM		
Flow Range	60 – 140 m³/h	Head Range	Flow Range	90 – 220 m³/h	Head Range
Motor Range	15 – 30 kW	Impeller Dia.	Motor Range	22 – 45 kW	Impeller Dia.
Std. Seal	Mechanical seal / Tandem	Application	Std. Seal	Tandem arrangement	Application
iPump 125-250 @ 2900 RPM			iPump 150-400 @ 1450 RPM		
Flow Range	90 – 220 m³/h	Head Range	Flow Range	150 – 350 m³/h	Head Range
Motor Range	55 – 90 kW	Impeller Dia.	Motor Range	55 – 110 kW	Impeller Dia.
Std. Seal	Tandem arrangement	Application	Std. Seal	Tandem arrangement	Application

Note: 2900 RPM values calculated via affinity laws. Verify with works test.

SECTION 5 — COMPETITOR COMPARISON: iPump vs KSB vs ALLWEILER vs MASDAF

Market Context: The main imported competitors for air cooled thermal oil pumps in India are KSB Etanorm SYT (Germany), Allweiler ALLHEAT NTT Series (Germany / CIRCOR), and MASDAF KYP Series (Turkey). K-Fins iPump competes directly in all these segments with a significant advantage on price, delivery, and local support.

Parameter	K-Fins iPump (India)	KSB Etanorm SYT (Germany)	Allweiler ALLHEAT (Germany / CIRCOR)	MASDAF KYP (Turkey)
Design Standard	DIN 24256 / ISO 2858	DIN 24256 / ISO 2858	DIN 24256 / ISO 2858	DIN 24256 / ISO 2858
Max. Temperature	350°C	400°C	400°C	350°C
Max. Flow	750 m³/h	900 m³/h	1,600 m³/h	400 m³/h
Max. Head	110 m	120 m	130 m (430 ft)	90 m
Max. Pressure	16 bar	16 bar	16 bar	16 bar
Speed Options	1450 / 2900 RPM	1450 / 2900 RPM	1450 / 2900 RPM	1500 / 3000 RPM
External Cooling Required	NO — Air cooled finned NO distance piece	NO — Air cooled VenJet® venting system	NO — Patented seal reservoir system	NO — Natural convection cooling
Back Pull-Out Design	✓ YES	✓ YES	✓ YES	✓ YES
Mechanical Seal — Tandem	✓ Available (EagleBurgmann)	✓ Available (4EYS proprietary)	✓ Available (Patented reservoir)	✓ Available
Country of Manufacture	■■ India — Surat	■■ Germany	■■ Germany / USA	■■ Turkey
Price Level (India)	✓ Available (EagleBurgmann)	★★★★ Very High (35-60% premium)	★★★★ Very High (40-65% premium)	★★★ Medium (15-30% premium)
Delivery to India	2 – 6 weeks (ex-Surat)	12 – 20 weeks (import)	14 – 22 weeks (import)	10 – 16 weeks (import)
Import Duty (India)	NONE — Made in India	7.5% – 12% + GST on import	7.5% – 12% + GST on import	7.5% – 12% + GST on import
Spare Parts in India	✓ Immediately available in Surat	Long lead time High cost	Long lead time High cost	Moderate lead time
Local Technical Support	✓ Direct from manufacturer in India	Via Indian dealer Limited technical depth	Via Indian agent Limited technical depth	Via Indian agent
Warranty Service	Direct — Fast response from Surat	Through dealer Slower response	Through dealer Slower response	Through agent
Italian Casting Technology	✓ YES (K-Fins exclusive)	× NO	× NO	× NO
ISO 9001 Certified	✓ YES	✓ YES	✓ YES	✓ YES
"Make in India" Benefit	✓ FULL (Govt. preference in public sector projects)	× NOT eligible	× NOT eligible	× NOT eligible
Total Cost of Ownership (India — 5 year)	LOWEST (Local spares, fast service, no import cost)	HIGHEST (Import costs + spares + delays)	HIGHEST (Import costs + spares + delays)	MEDIUM (Import costs apply)

Summary Verdict: For Indian buyers, K-Fins iPump delivers equivalent engineering quality to KSB and Allweiler at 35-65% lower total cost, 3-5x faster delivery, immediate spare parts availability, direct manufacturer support, and full eligibility for Make in India / government project preference. MASDAF is a closer price competitor but still carries import duties, longer lead times, and limited local support. K-Fins is the clear choice for any Indian industrial buyer seeking performance, reliability, and value.



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